Assignment 2 | End-to-end classification and deployment

Student	Apeksha Manoj Patel
BITS ID	2026AJ05790

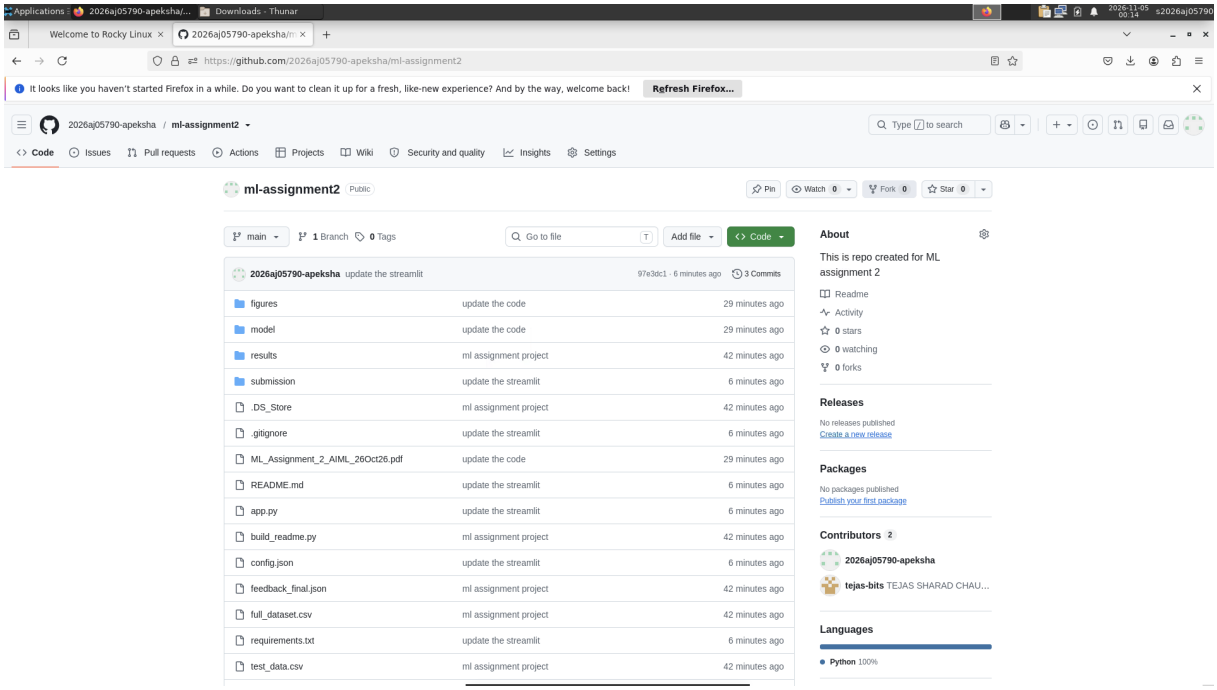
1. GitHub Repository Link

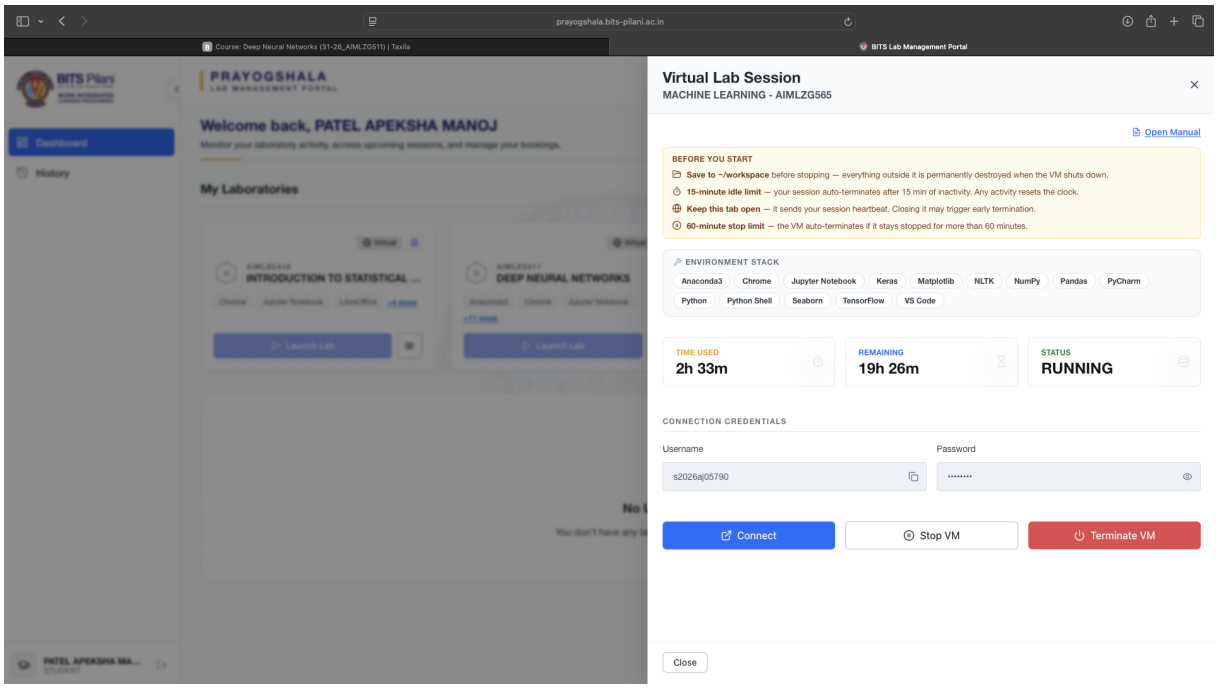
https://github.com/2026aj05790-apeksha/ml-assignment2

2. Live Streamlit App Link

https://ml-assignment2-bxkybtx2fackxbtb4bh9vk.streamlit.app

3. BITS Virtual Lab usage screenshots




prayogshala.bits-pilani.ac.in
Course: Deep Neural Networks (S1-26_AIMLZG511) | Taxila
BITS Lab Management Portal
PRAYOGSHALA
LAB MANAGEMENT PORTAL
Dashboard
History
Welcome back, PATEL APEKSHA MANOJ
Monitor your laboratory activity, access upcoming sessions, and manage your bookings.
My Laboratories
INTRODUCTION TO STATISTICAL ...
DEEP NEURAL NETWORKS
Launch Lab
Virtual Lab Session
MACHINE LEARNING - AIMLZG565
Open Manual
BEFORE YOU START
Save to ~/workspace before stopping — everything outside it is permanently destroyed when the VM shuts down.
15-minute idle limit — your session auto-terminates after 15 min of inactivity. Any activity resets the clock.
Keep this tab open — it sends your session heartbeat. Closing it may trigger early termination.
60-minute stop limit — the VM auto-terminates if it stays stopped for more than 60 minutes.
ENVIRONMENT STACK
Anaconda3
Chrome
Jupyter Notebook
Keras
Matplotlib
NLTK
NumPy
Pandas
PyCharm
Python
Python Shell
Seaborn
TensorFlow
VS Code
TIME USED
2h 33m
REMAINING
19h 26m
STATUS
RUNNING
CONNECTION CREDENTIALS
Username
Password
s2026aj05790
Connect
Stop VM
Terminate VM
Close

4. README Content

■ Breast-Cancer Diagnosis – Five-Classifier Comparison & Streamlit Deployment

BITS Pilani WILP · M.Tech (AIML) · Machine Learning · Assignment 2

a. Problem statement

Early, accurate discrimination between **malignant** and **benign** breast tumours from fine-needle-aspirate (FNA) cell-nucleus measurements reduces unnecessary biopsies and prevents missed cancers.

The task is a **binary classification** problem: build, evaluate and compare five classifiers (Logistic Regression, Decision Tree, KNN, Naive Bayes, SVM) on one dataset using six metrics

(Accuracy, AUC, Precision, Recall, F1, MCC), then deploy an interactive Streamlit app that evaluates the *saved* models on an uploaded test CSV without retraining.

b. Dataset description

Item	Detail
Source	Breast Cancer Wisconsin (Diagnostic) – UCI Machine Learning Repository (Wolberg, Street & Mangasarian, 1995), loaded via `sklearn.datasets.load_breast_cancer`
Instances	569 (train 455 / test 114, stratified 80:20, seed 42)
Features	30 numeric (>= 12 required): 10 nucleus measurements × {mean, standard error, worst}
Target	`diagnosis` – 0 = malignant (212, 37.3 %), 1 = benign (357, 62.7 %)
Class balance	Moderately imbalanced (1 : 1.68) → MCC and recall reported alongside accuracy
Missing values	None
Pre-processing	`StandardScaler` inside each sklearn `Pipeline` (fit on training data only – no leakage); the saved `.joblib` therefore contains the complete preprocessing

The 10 base measurements: radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, fractal dimension.

Many are strongly collinear (radius–perimeter–area r > 0.95), which matters for Naive Bayes (see observations).

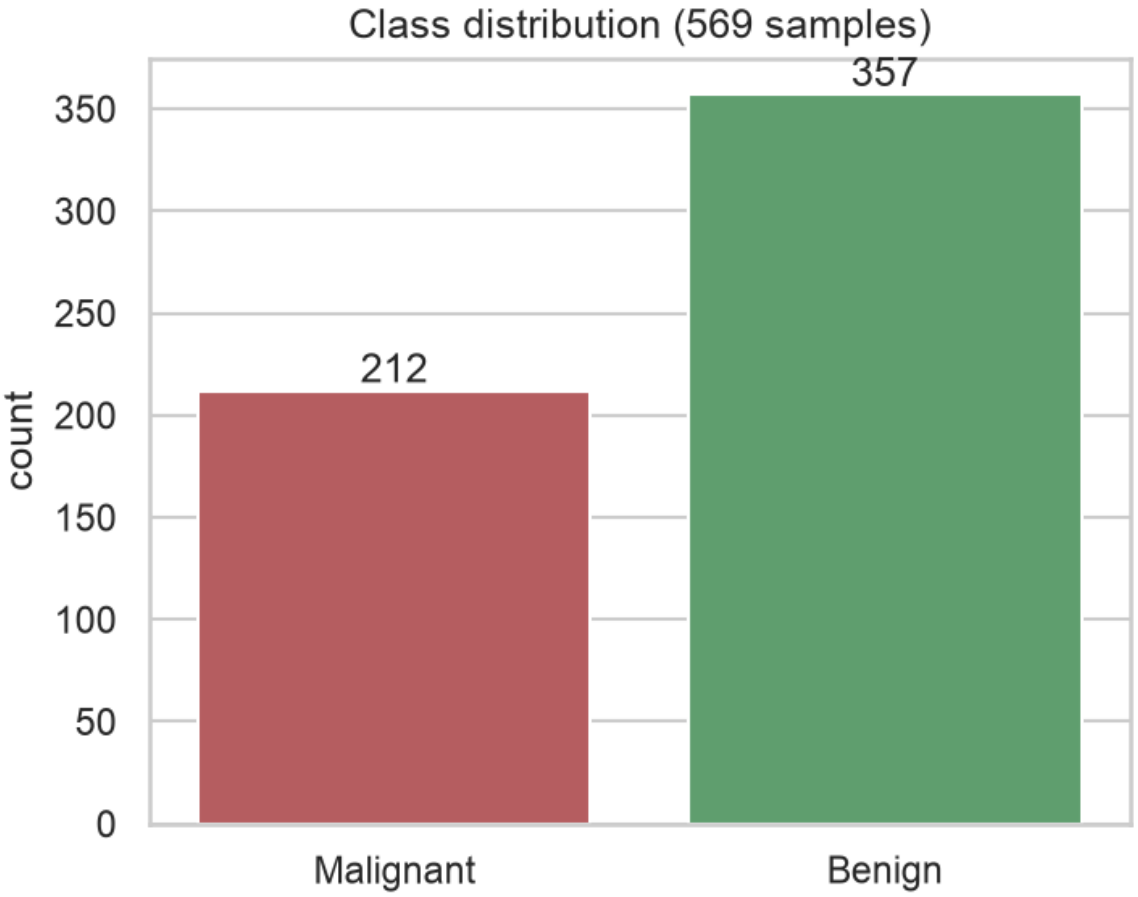


Figure 1 – Class distribution

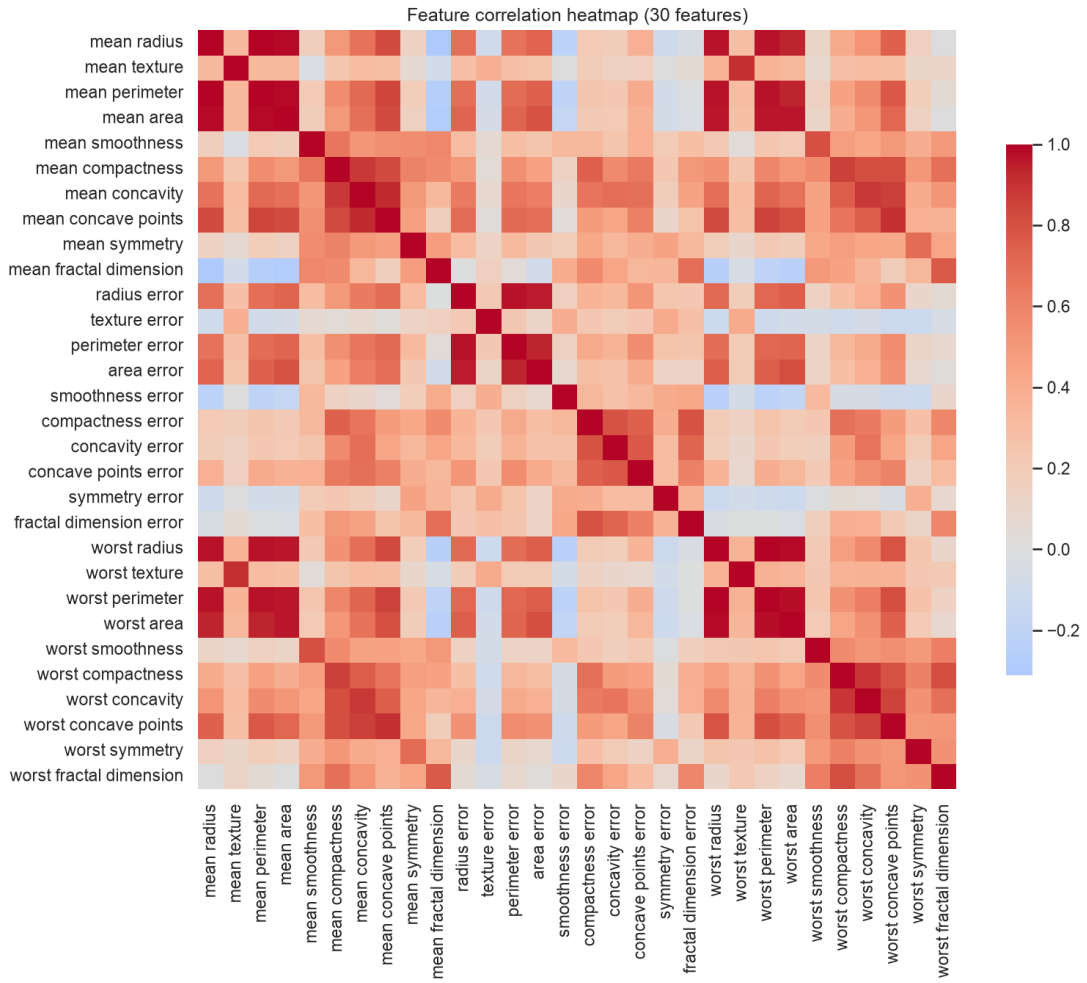


Figure 2 – Feature correlation heatmap: blocks of highly correlated size features

c. GitHub Repository Link

https://github.com/2026aj05790-apeksha/ml-assignment2

```
project-folder/
|-- src/                    # application, training, and README-generation code
|-- artifacts/              # generated data, models, metrics, and figures
|-- test_data.csv           # rubric-required hold-out test data mirror
|-- model/                  # rubric-required saved-model mirror
|-- problem_statement/      # original assignment PDF
|-- submission/             # final PDF and BITS Lab screenshots
|-- config/                 # student, link, and submission metadata
|-- reports/                # grader feedback and reports
`-- requirements.txt
```

Live Streamlit app: https://ml-assignment2-bxkybtx2fackxbtb4bh9vk.streamlit.app

How to run / reproduce

```
pip install -r requirements.txt
python train.py           # retrains all models and regenerates artifacts/
streamlit run app.py      # launches the evaluation app
```

d. Models used

All five models are trained on the **same** 455-row training split and evaluated on the **same** 114-row hold-out test set.

Hyper-parameters: LR (C=1), Decision Tree (max_depth=5, min_samples_leaf=3), KNN (k=7, distance-weighted), Gaussian NB (defaults), SVM (RBF, C=1, gamma='scale').

Comparison table (test set)

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.9825	0.9954	0.9861	0.9861	0.9861	0.9623
Decision Tree	0.9035	0.9373	0.9420	0.9028	0.9220	0.7969
KNN	0.9737	0.9884	0.9600	1.0000	0.9796	0.9442
Naive Bayes	0.9298	0.9868	0.9444	0.9444	0.9444	0.8492
Support Vector Machine	0.9825	0.9950	0.9861	0.9861	0.9861	0.9623

Generalisation check (5-fold stratified CV on training data vs train vs test)

ML Model Name	Train Acc	5-fold CV Acc	Test Acc
Logistic Regression	0.9890	0.978 ± 0.010	0.9825
Decision Tree	0.9802	0.916 ± 0.019	0.9035
KNN	1.0000	0.963 ± 0.020	0.9737
Naive Bayes	0.9385	0.934 ± 0.029	0.9298
Support Vector Machine	0.9824	0.969 ± 0.015	0.9825

Additional diagnostic metrics

The required six metrics are shown above. The following measures provide a more complete view of class-specific and imbalance-aware performance. Here, class 1 (benign) is treated as the positive class; therefore specificity is the ability to identify malignant cases (class 0), while FNR is the proportion of benign cases predicted as malignant.

ML Model Name	Balanced Accuracy	Specificity	NPV	FNR
Logistic Regression	0.9812	0.9762	0.9762	0.0139
Decision Tree	0.9038	0.9048	0.8444	0.0972
KNN	0.9643	0.9286	1.0000	0.0000
Naive Bayes	0.9246	0.9048	0.9048	0.0556
Support Vector Machine	0.9812	0.9762	0.9762	0.0139

Confusion-matrix counts

ML Model Name	TN	FP	FN	TP
Logistic Regression	41	1	1	71
Decision Tree	38	4	7	65
KNN	39	3	0	72
Naive Bayes	38	4	4	68
Support Vector Machine	41	1	1	71

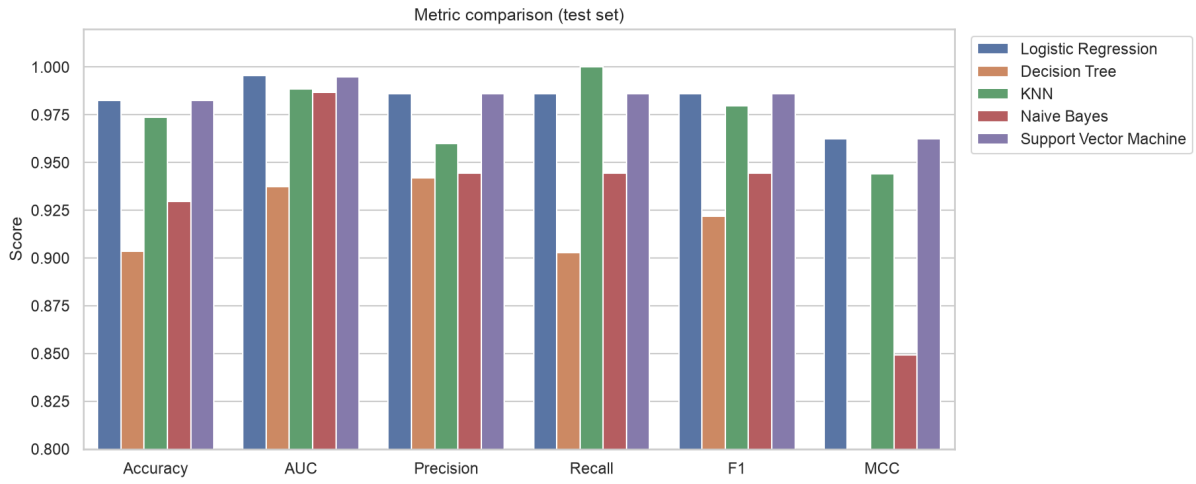


Figure 3 – Metric comparison across models

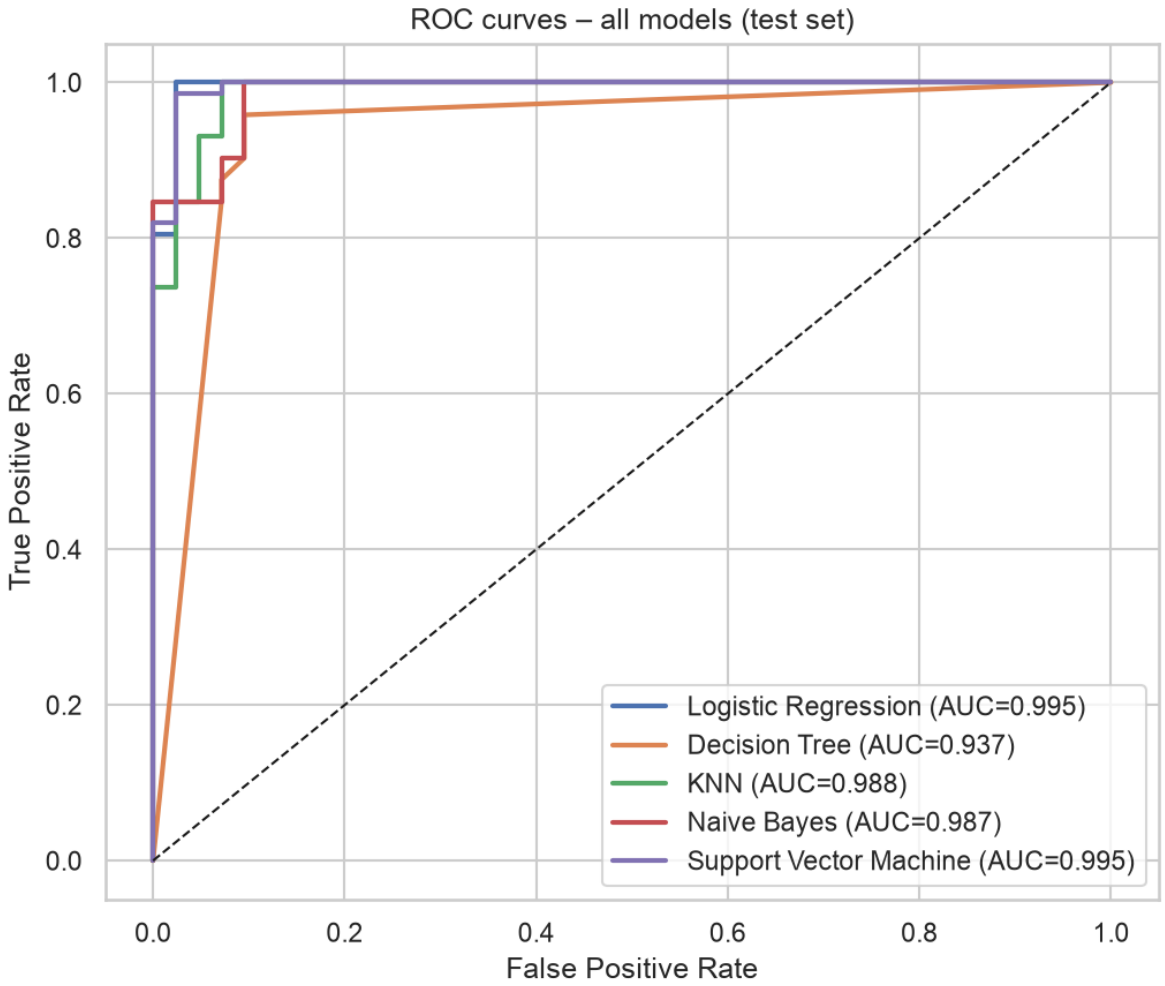


Figure 4 – ROC curves (test set)

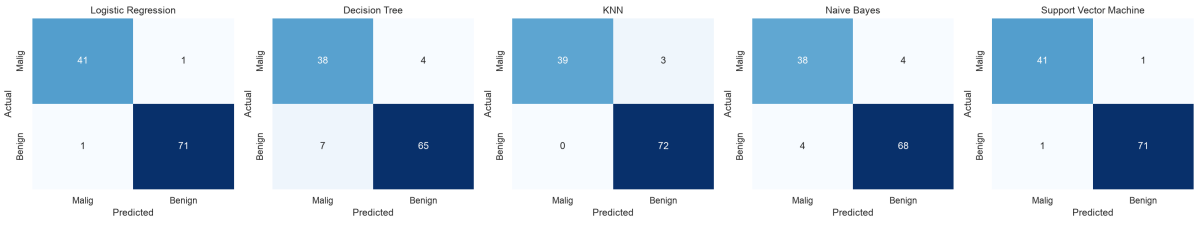


Figure 5 – Confusion matrices (class 0 = malignant)

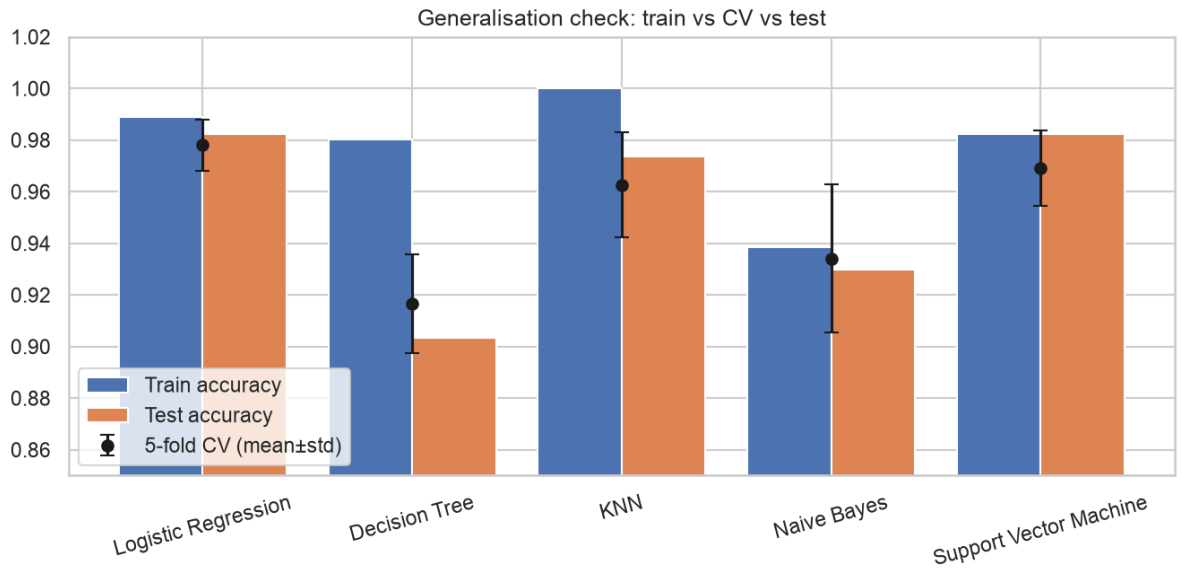


Figure 6 – Train vs CV vs test accuracy

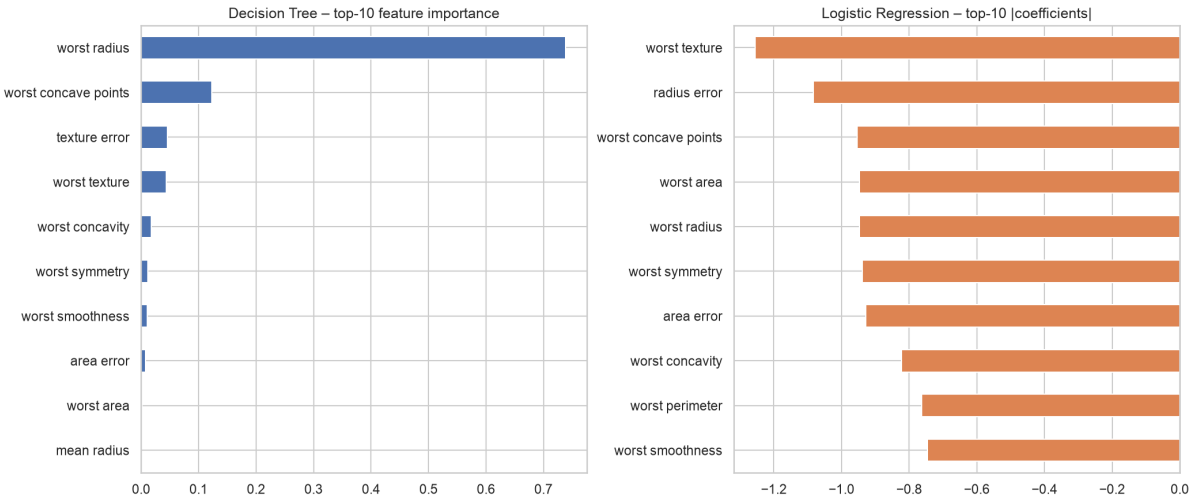


Figure 7 – Feature importance (Decision Tree) and coefficients (Logistic Regression)

Observations on model performance

ML Model Name	Observation about model performance
Logistic Regression	Reaches 98.25% accuracy, AUC 0.9954 and MCC 0.962 (tied best), with only 2 errors out of 114 test rows. This works because the malignant/benign classes are almost linearly separable in standardised feature space, so a linear boundary has low bias and, with only 30 inputs, low variance. Train accuracy (0.989) is close to CV (0.978 ± 0.010), indicating no overfitting. Coefficients are directly interpretable (see feature-importance figure).
Decision Tree	Weakest model: accuracy 90.35%, AUC 0.937, MCC 0.797. Train accuracy (0.980) far exceeds test accuracy, a classic high-variance overfitting signature even with max_depth=5 and min_samples_leaf=3, because axis-aligned splits must approximate a diagonal boundary with many small steps and 455 training rows are too few to stabilise them. It missed 4 malignant cases (recall 0.903), clinically the worst outcome. Its coarse probability estimates also pull AUC down. Benefit: fully transparent rules.
KNN	Accuracy 97.37% and perfect recall (1.000) with 3 missed malignant cases and 0 false positives, so precision (0.960) is the cost. Distance-weighted k=7 on standardised features works because similar tumours cluster in feature space; without scaling it would be dominated by area/perimeter features. Train accuracy is 1.000 (distance weighting memorises training points) while CV is 0.963 ± 0.020, so the CV standard deviation is the largest of the distance-based models and prediction is slower since it stores all data.
Naive Bayes	Accuracy 92.98%, MCC 0.849 but a strong AUC of 0.9868, meaning it ranks cases well yet its 0.5 cut-off is poorly calibrated. The independence assumption is violated here because radius, perimeter and area (and their 'worst' versions) are correlated above 0.95 (see heatmap); correlated evidence is counted multiple times, making posteriors overconfident. Train (0.939) and test accuracies are similar, so it suffers from bias rather than variance. It is the fastest model and needs no tuning.
Support Vector Machine	Ties with Logistic Regression on every test metric (accuracy 98.25%, AUC 0.9950, MCC 0.962). The RBF kernel maximises the margin and handles any mild non-linearity, and scaling makes gamma='scale' meaningful. CV accuracy (0.969 ± 0.015) is slightly below logistic regression, suggesting the extra flexibility buys nothing on this nearly linear problem while costing training time and interpretability.
Overall, Winner for your dataset?	**Logistic Regression** (with SVM as a statistical tie). LR and SVM have identical test accuracy (98.25%), MCC (0.962) and recall, but LR has the higher 5-fold CV accuracy (0.978 ± 0.010 vs 0.969 ± 0.015), the highest AUC, trains in milliseconds and is interpretable. We rank by MCC and recall rather than accuracy alone, because MCC is robust to the 63/37 class imbalance and a false negative (missed malignancy) is the costliest error. KNN has the best recall but lower precision/MCC. With only 114 test rows, one sample moves accuracy by 0.88%, so LR vs SVM differences are not significant.

Error analysis

15 of 114 test samples are misclassified by at least one model and **1** by all five – these are genuinely borderline tumours whose measurements overlap the other class.

Most of the accuracy gap between models therefore comes from a small set of ambiguous cases, which is why MCC/recall and CV are used in addition to a single test split.

Streamlit application features

CSV upload (test data only) · model dropdown · six required metric cards · detailed per-class classification report · confusion matrix · specificity · balanced accuracy · ROC curve · adjustable decision threshold ·

side-by-side comparison of all five saved models on the uploaded data · prediction download · input validation (missing columns / non-binary target).

Models are loaded with `joblib`; **no training happens in the app**.

Limitations

• Single public dataset with 569 rows; a 114-row test set gives wide confidence intervals (one sample ≈ 0.88 % accuracy), so LR vs SVM differences are not statistically significant.

• Hyper-parameters were set by reasoned defaults and light CV, not an exhaustive search; data come from one institution, so external generalisation is unverified.

• Models are decision-support tools, not diagnostic devices.

Future work

• Nested cross-validation with grid/Bayesian search, probability calibration (Platt/isotonic) – especially for Naive Bayes.

• Ensembles (Random Forest, Gradient Boosting, stacking), SHAP explanations, external validation on another cohort.

• Cost-sensitive threshold selection to explicitly minimise false negatives.